

Occlusion Robust Part-aware Object Classification through Part Attention and Redundant Features Suppression

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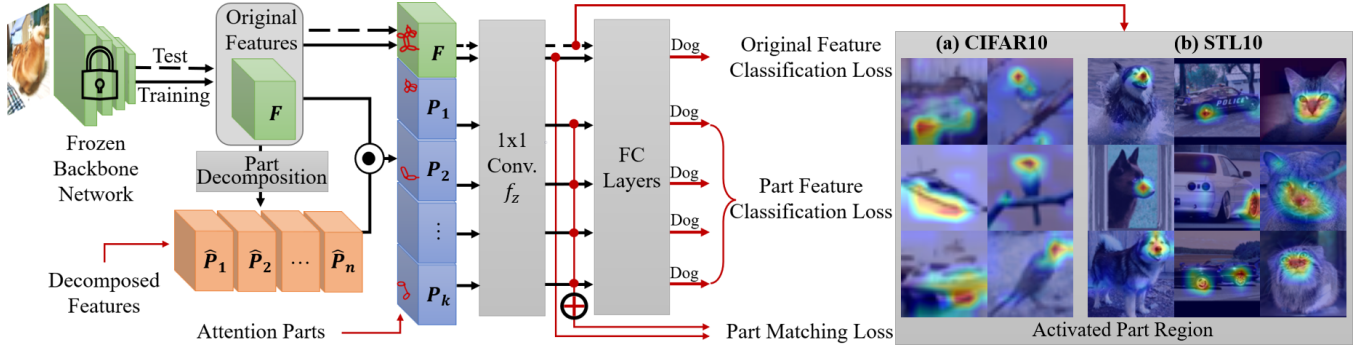


Figure 1: Our proposed method: In training, original object features are decomposed into part features of the object, and both original features and the decomposed part features are used to train f_z and FC Layers for object classification. In test, only original object features are used for the classification. Each column of (a) and (b) corresponds to the output of the same channel of f_z and represents activated part region.

ABSTRACT

In recent studies, object classification with deep convolutional neural networks has shown poor generalization with occluded objects due to the large variation of occlusion situations. We propose a part-aware deep learning approach for occlusion robust object classification. To demonstrate the robustness of the method to unseen occlusion, we train our network without occluded object samples in training and test it with diverse occlusion samples. Proposed method shows improved classification performance on CIFAR10, STL10, and vehicles from PASCAL3D+ datasets.

CCS CONCEPTS

• Computing methodologies → Object recognition.

KEYWORDS

Occlusion, Object Classification, Non-negative Matrix Factorization

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1 INTRODUCTION

Although deep convolutional neural networks (DCNNs) have been tremendously improved in their classification performance, recent researches show that DCNNs are not robust to the classification of partially occluded objects as summarized in [Kortylewski et al. 2020a]. Difficulties of occlusion include the large variation of size and shape of the occluded region (and exposed region) and varying appearance of occluders. DCNNs rely on the set of most discriminative features that could be included in the occluded region. Training sample occlusion augmentation cannot simulate all such variations of occlusions. [Kortylewski et al. 2020a,b] cluster object features and compare them with pre-defined features of occluders to be robust to partial occlusion through compositional neural networks. They propose clustering-based feature modelings of object parts that have limitations in the case of larger intra-class variation of samples than inter-class variation. Furthermore, it is difficult to define the features of numerous occluder classes. In this work, we concentrate on part decomposed descriptions of objects regardless of occluder types and characteristics. Inspired by [Collins et al. 2018], we propose a method to decompose objects features into their part features using part attention and redundant features suppression schemes and classify objects with the combination of object-part features. Our part-aware object classification enables our network to classify occluded objects with a mixture of remaining parts.

2 PROPOSED METHOD

Figure 1 shows the overall framework of our method. We train our network without occluded object samples and test it with occluded object samples. Proposed method decomposes features from

a backbone network into part features of an object. And then, we give attention to object parts and suppress redundant features by masking with the decomposed part features. Finally, FC layers are trained based on both original object features and part features of the object for object classification, which alleviates the dependency of the network on the most discriminative features. FC layers include two fully connected layers with ReLU activations.

Let $F \in \mathbb{R}^{(h \cdot w) \times c}$ denotes a feature map of an input from a backbone network, where $\mathbb{R}$ indicates the set of non-negative real numbers, and h , w and c refer to height, width and channel of the feature map, respectively. Through Non-negative Matrix Factorization(NMF), we factorize F as $\hat{F}_k = WH$, $W, H \geq 0$, where $\hat{F}_k$ is a row-rank feature map of a user-defined rank k , $W \in \mathbb{R}^{(h \cdot w) \times k}$, and $H \in \mathbb{R}^{k \times c}$. Each row vector of H represents the center of an object-part feature and each column vector of W describes an activated region of object-part, as explained in [Collins et al. 2018]. Given the factorized features, attention parts P_n s are computed as follows.

$$\hat{P}_n = \frac{w_n}{\|w_n\|_2} \frac{h_n^T}{\|h_n^T\|_2}, \quad (1)$$

$$P_n = F \odot \hat{P}_n, (1 \leq n \leq k),$$

where $\hat{P}_n \in \mathbb{R}^{(h \cdot w) \times c}$ denotes part coefficient (decomposed features), $w_n \in \mathbb{R}^{(h \cdot w) \times 1}$ and $h_n^T \in \mathbb{R}^{1 \times c}$ indicates column and row vector of W and H respectively, and $\odot$ is element-wise multiplication. It is possible to assign different rank to different classes. However, since it is impossible to know proper combination of object-parts(rank) of an arbitrary test sample, we add 1x1 convolutional layers f_z to decide part features to give attention and redundant features to suppress such as backgrounds. f_z consists of three 1x1 convolutional layers with ReLU activations. Finally, FC layers are trained to classify both $f_z(F)$ and $f_z(P_n)$ s to recognize an input as a union of all object-parts. We encode this into loss regularization terms. Overall loss function is formulated as follows.

$$\mathcal{L} = \mathcal{L}_{F_{cls}} + \alpha \mathcal{L}_{P_{cls}} + \beta \mathcal{L}_{pm},$$

$$\mathcal{L}_{pm} = \left\| f_z(F) - \left(\sum_i^k f_z(P_i) / k \right) \right\|^2, \quad (2)$$

where $\mathcal{L}_{F_{cls}}$ and $\mathcal{L}_{P_{cls}}$ denote the object classification loss(cross-entropy) of original features and part features respectively, $\mathcal{L}_{pm}$ indicates feature matching loss, i is the index of the parts, and α and β are weights of loss regularization term. $\mathcal{L}_{pm}$ induces f_z to detect whether F has P_n s by comparing original features and the sum of part features. In test, FC layers classify $f_z(F)$.

3 EXPERIMENTAL EVALUATIONS

Experimental evaluations are performed on three benchmark datasets; CIFAR10, STL10 and vehicles from PASCAL3D+. We use VGG16 with batch normalization for backbone networks in CIFAR10, STL10 and PASCAL3D+ test, and ResNet34 with batch normalization in

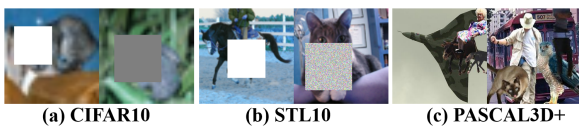


Figure 2: Occlusion samples used for evaluation test

Table 1: Classification results(%) on CIFAR10 and STL10: Original denotes test with non-occluded samples. All tests are repeated three times and averaged. (·) indicates the index of layer used for feature decomposition.

	CIFAR10		STL10	
	Original	Occlusion	Original	Occlusion
VGG16bn	88.72	73.76	94.23	86.93
VGG16bn + 1x1 Conv (40)	91.14	76.54	94.62	85.82
VGG16bn + 1x1 Conv (37)	91.71	78.79	94.21	87.06
OURS (40)	91.31	78.72	94.47	87.88
OURS (37)	91.81	81.58	93.88	88.82
ResNet34bn	89.74	59.39	95.7	86.41
OURS	93.55	77.18	94.25	86.96

Table 2: Classification results(%) on PASCAL3D+ occluded by natural objects: CompNet¹ and CompNet²⁵⁶ are trained with batch size of 1 (in author's code) and 256, respectively.

Occ.Area	0	Level1: 20-40%	Level2: 40-60%	Level3: 60-80%
CompNet ¹ [Kortylewski et al. 2020a]	98.24	95.73	90.62	80.58
VGG16bn	99.47	93.29	82.26	69.40
CompNet ²⁵⁶ [Kortylewski et al. 2020a]	97.99	91.63	74.63	57.6
OURS	99.33	95.62	88.55	76.15

CIFAR10, STL10 test. We randomly occlude 25-50% area of each sample of CIFAR10 and STL10 with white, gray and noise boxes. It is possible to assign different rank to different classes for feature decomposition but we empirically assign rank4 and rank5 for all classes in CIFAR10 and STL10, respectively. As shown in table 1, our method outperforms baselines. We compare VGG16bn with 1x1 convolutional layers. Since 1x1 convolutional filters only consider single pixel, network extracts independent features of each pixel. It is helpful to extract such independent features in partial occlusion cases diminishing the correlation between adjacent pixels.

For the evaluation with naturally occluded objects, we follow the existing evaluation benchmark of PASCAL3D+ occluded by natural objects introduced in [Kortylewski et al. 2020a,b] (Figure 2). We use officially distributed code by [Kortylewski et al. 2020a]. As summarized in table 2, CompNet¹ [Kortylewski et al. 2020a] based on VGG16 outperforms other methods. However, its performance decreases by 22% with batch size of 256(CompNet²⁵⁶ [Kortylewski et al. 2020a]), which is the same training condition as ours. As described in Figure 1(a) and (b), same parts of a class is activated in a channel of the last 1x1 convolutional layer. This implies that our 1x1 convolutional layers properly detect part features in the tests.

4 CONCLUSION

We propose occlusion robust part-aware object classification through part attention and redundant features suppression. Our method robustly classify occluded objects in our evaluations on three benchmarks without training with occluded object samples.

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